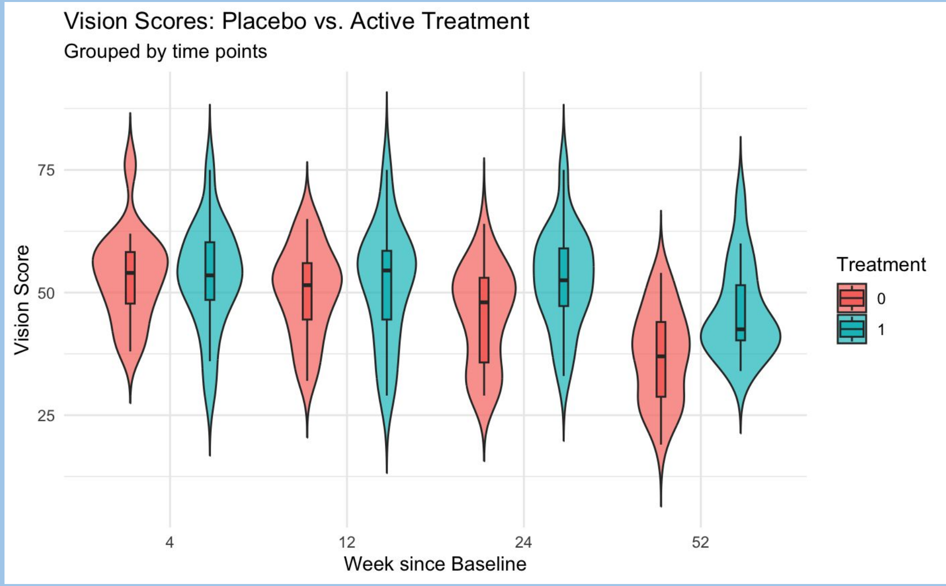
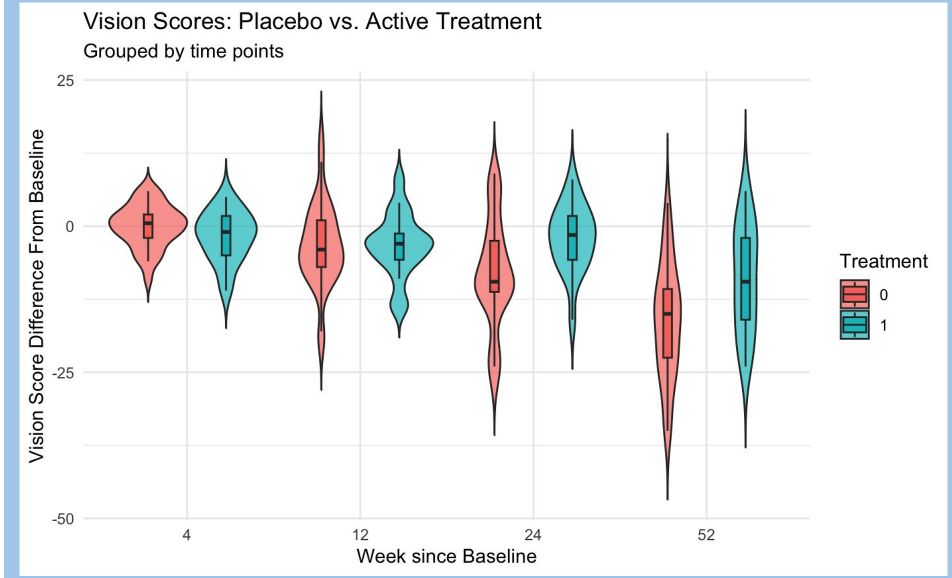
Age Related Macular Degeneration

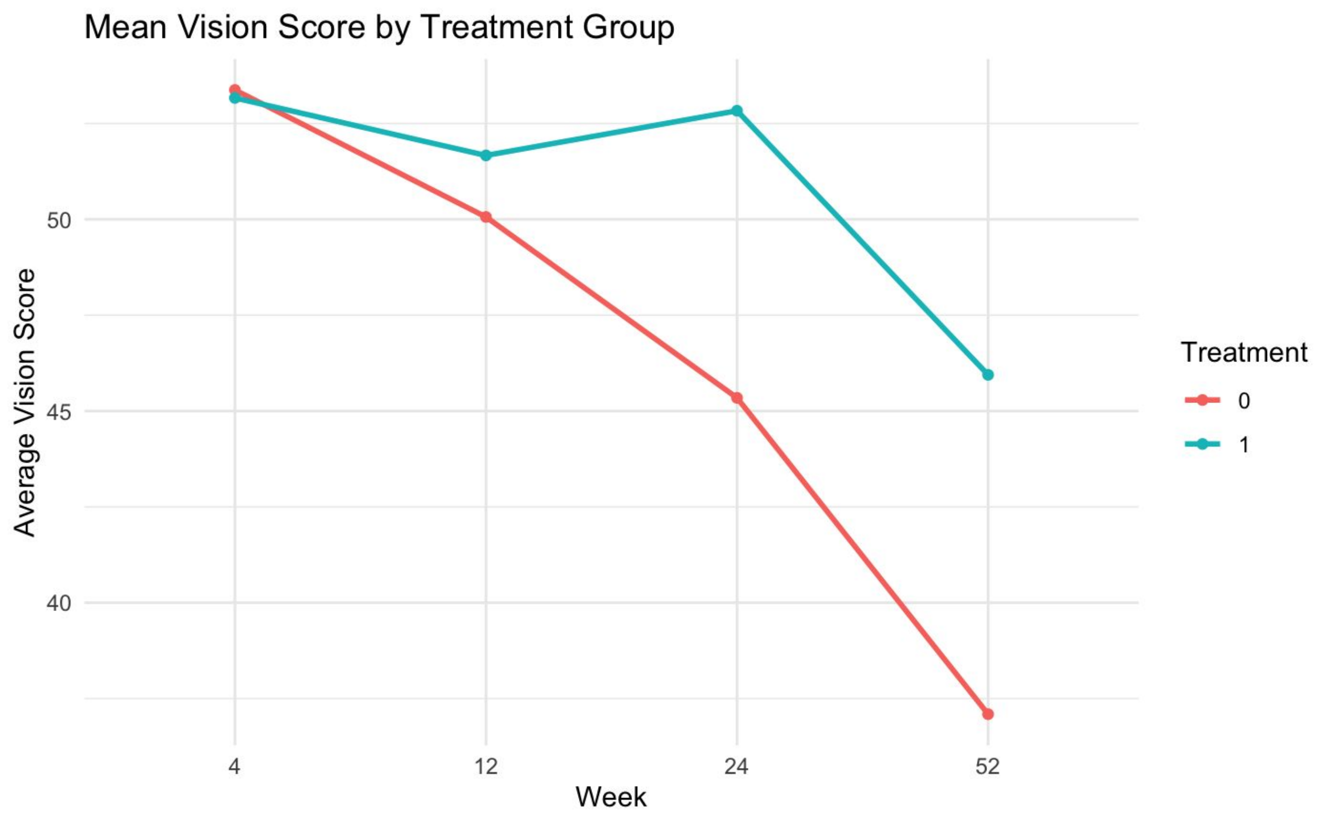
Andrea Reed and Kimberly Williams

EDA





Both groups trend down, but the interferon-α appears to decrease slower (even when accounting for the baseline).



The Mean Vision Score for Treatment 1 (interferon-α) appears to be higher as time progresses.

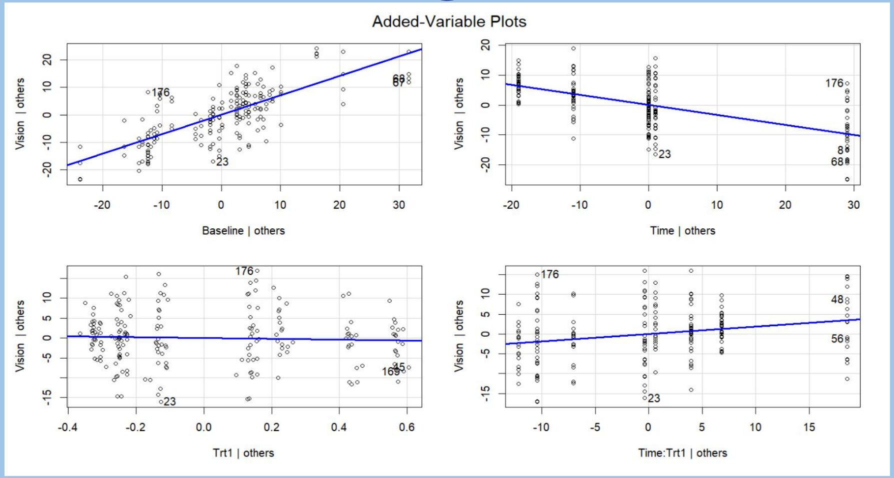
Modeling

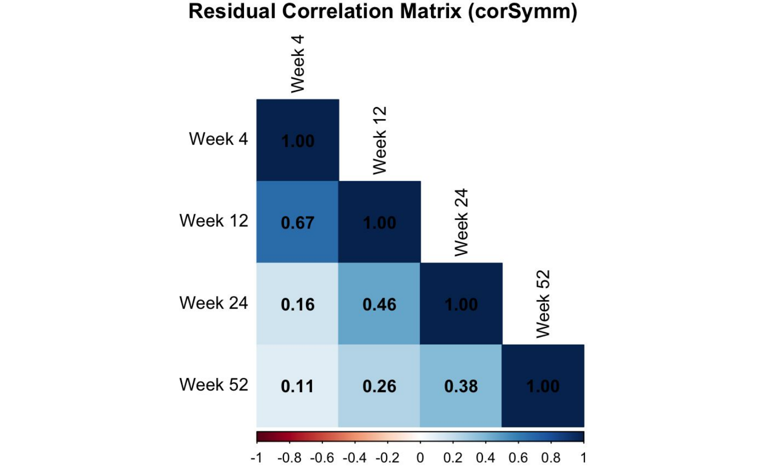
We used a block linear model with interaction terms to account for the repeated measurements on the same patients.

Vision ~ Baseline + Time + Trt +Time:Trt

Vision →The patient's vision score at time of follow up (higher scores mean better vision)
Baseline →The patient's baseline vision score
Time →The week since the baseline vision was measured
Trt → Did the patient receive the treatment (Trt=1=Yes)
Time:Trt → We include the interaction term based on ANOVA with a p-val of 0.0526

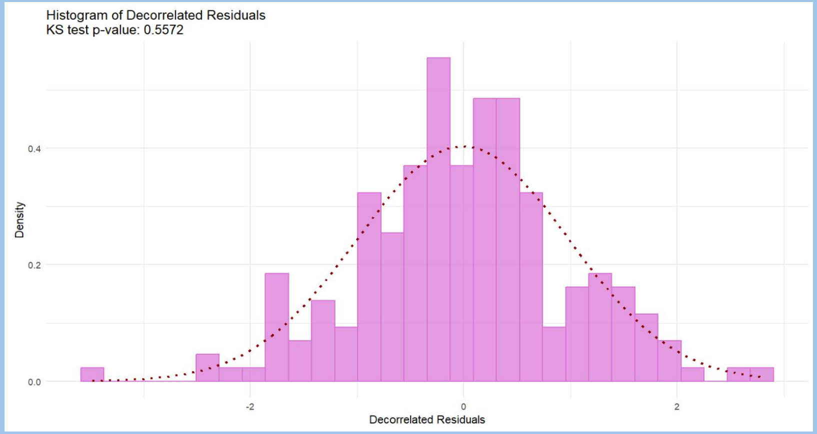
Validating Model Assumptions



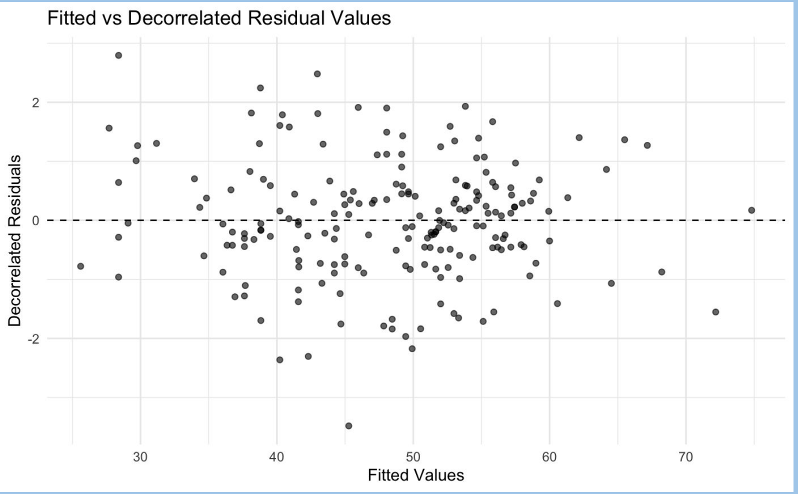


The data is linear, but the residuals are too correlated (since we’re looking at individuals over 5 different time points: baseline, week 4, 12, 24, and 52)

Validating Model Assumptions



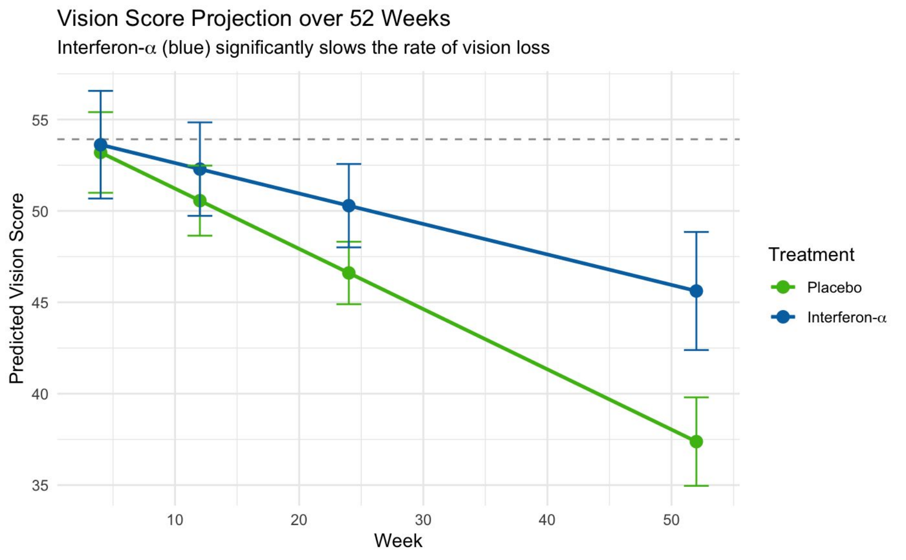
	res_4	res_12	res_24	res_52
res_4	1.00000000	0.07503945	0.01603990	0.05999283
res_12	0.07503945	1.00000000	0.07083154	0.07595846
res_24	0.01603990	0.07083154	1.00000000	0.08105748
res_52	0.05999283	0.07595846	0.08105748	1.00000000



64% of the variability in the data can be explained by our model. (Pseudo R^2)

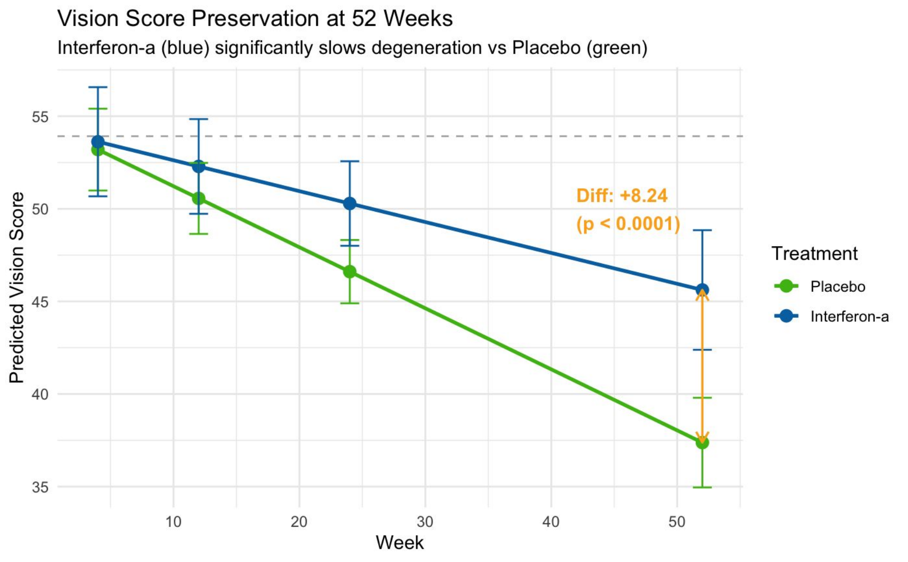
Does interferon-α stop vision loss over time?

Because the confidence interval includes 0, we know that it does not stop macular degeneration over time. Interferon-α slows vision loss by about 0.163 per week compared to no treatment.



At 52 Weeks

After controlling for baseline vision, patients receiving interferon-α have significantly higher vision scores at 52 weeks compared to placebo ($p < 0.05$). The estimated difference is 8.24 letters, indicating a significant improvement. We are 95% confident the true improvement over no treatment lies within 4.2 to 12.2.



Thank you!